LESSON 8 • TERM 4 • LEVEL 1

# Graph Detectives

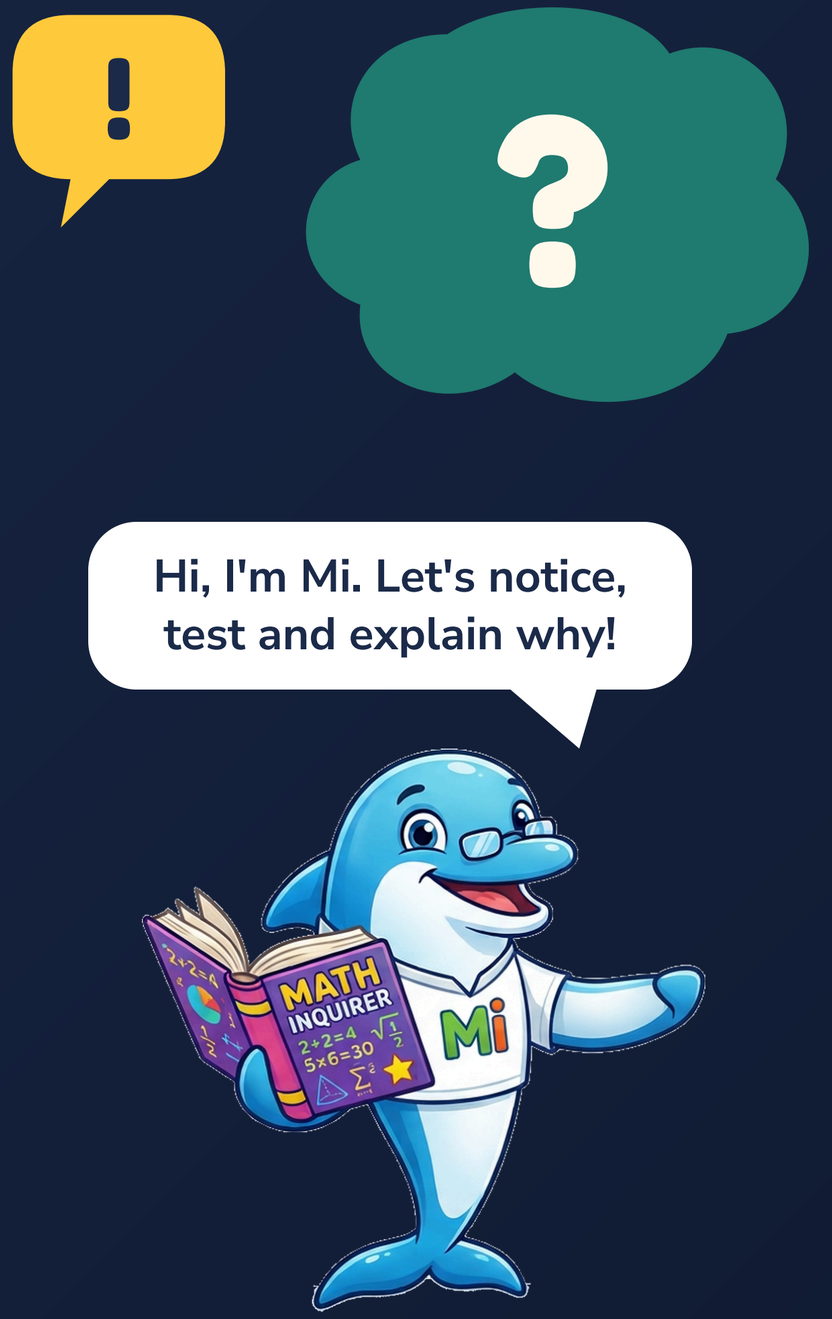
Read the key before you read the graph

## TODAY WE WILL

Read the key first.

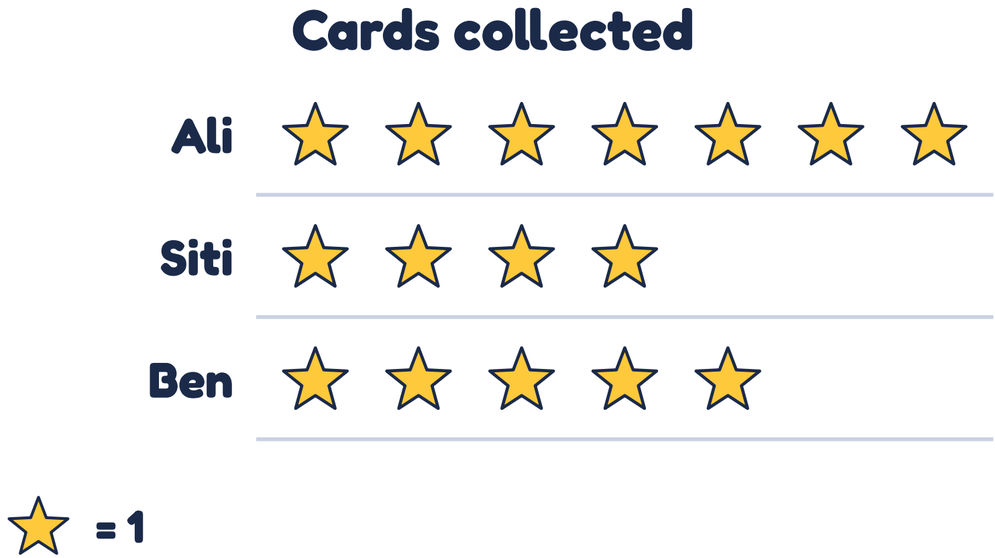
Find a missing bar.

Ask what a graph cannot tell us.



# Warm-up: read a graph

2–8 MIN




**Each picture is 1 card. Who has the most cards? How many more cards does Ali have than Ben?**

# Count in 2s and 3s

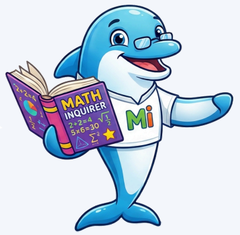
2–8 MIN







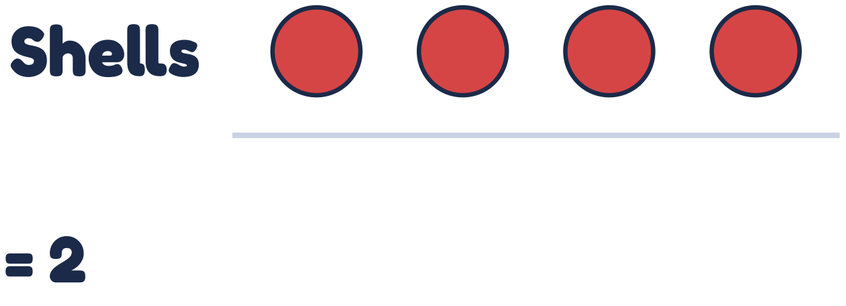
**Count in 2s. Then count in 3s. Say the missing numbers.**




Take your time!

# Pictures that stand for 2

2–8 MIN




Each picture stands for 2. Count in 2s along the row. How many shells are there?

# Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

## Stickers




**PUZZLE 1: anyone can try**

How many pictures are in Ravi's row?

Write the number.

**PUZZLE 2: think**

Each picture stands for 2 stickers.

How many stickers does Ravi have?

I know because ___.

# Explore: make pairs

12–23 MIN

1. Draw 8 dots in a row on your paper.
2. Draw a loop around every 2 dots.
3. Each loop is 1 picture. How many pictures?

**Now try 12 dots. How many pictures?**

# 4 pictures and 6 pictures

12–23 MIN

**8 dots make 4 pictures. 12 dots make 6 pictures. Each picture stands for 2 dots.**

# Explore: two graphs

12–23 MIN

## Graph A

Pens 

● = 1

## Graph B

Rulers 

● = 2

**Both rows have 4 pictures. Which row shows more things? How do you know?**

# Read the key first

12–23 MIN

**Graph A**

Pens 

● = 1

**Graph B**

Rulers 

● = 2

**Graph B shows more. Graph A shows 4 pens. Graph B shows 4 pictures of 2: 8 rulers.**

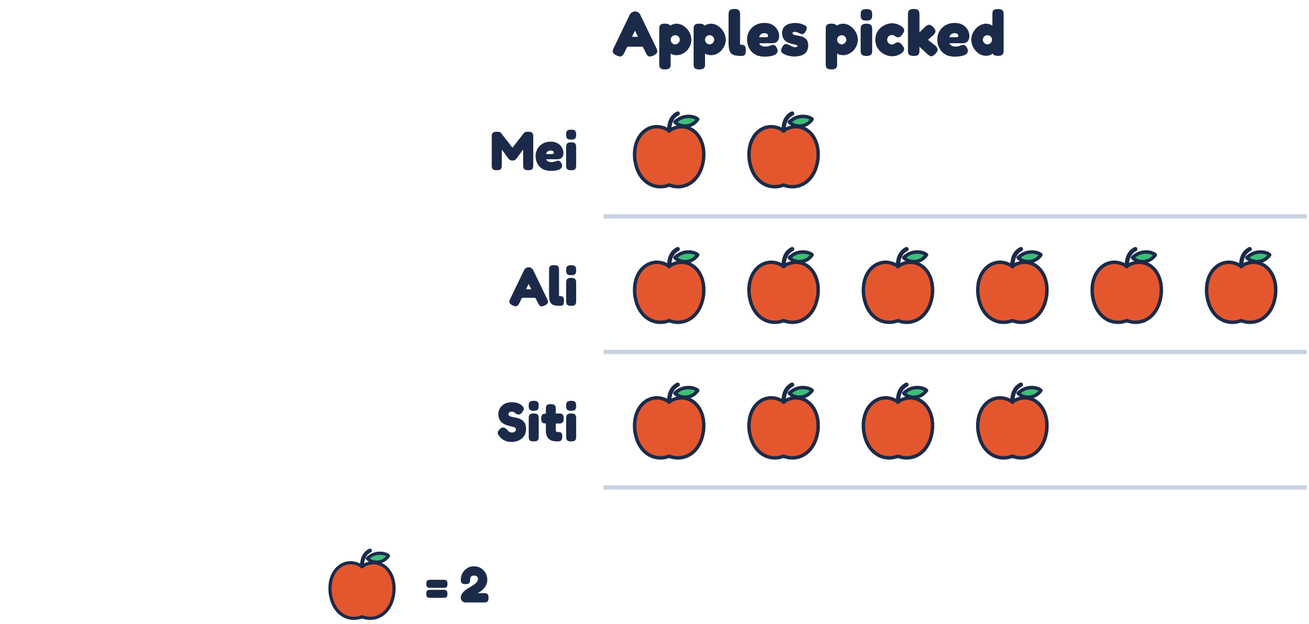
# What did we notice?

12–23 MIN

- What does the key tell us?
- Is the longest row always the most things?

# Read the key first

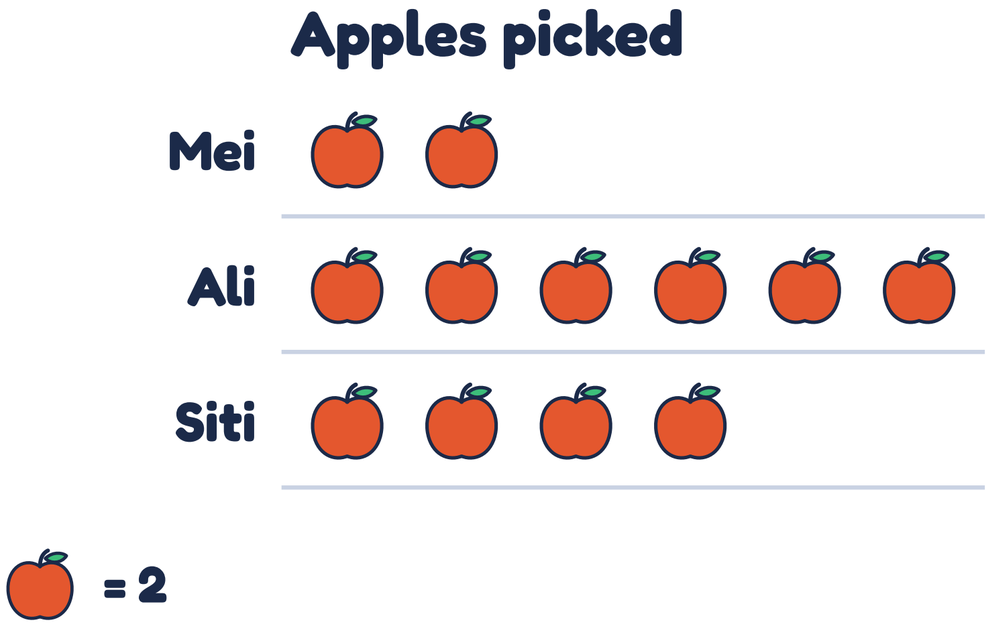
23–34 MIN




Read the key first. How many apples did Ali pick? How many more apples did Ali pick than Mei?

# Ali picked 12 apples

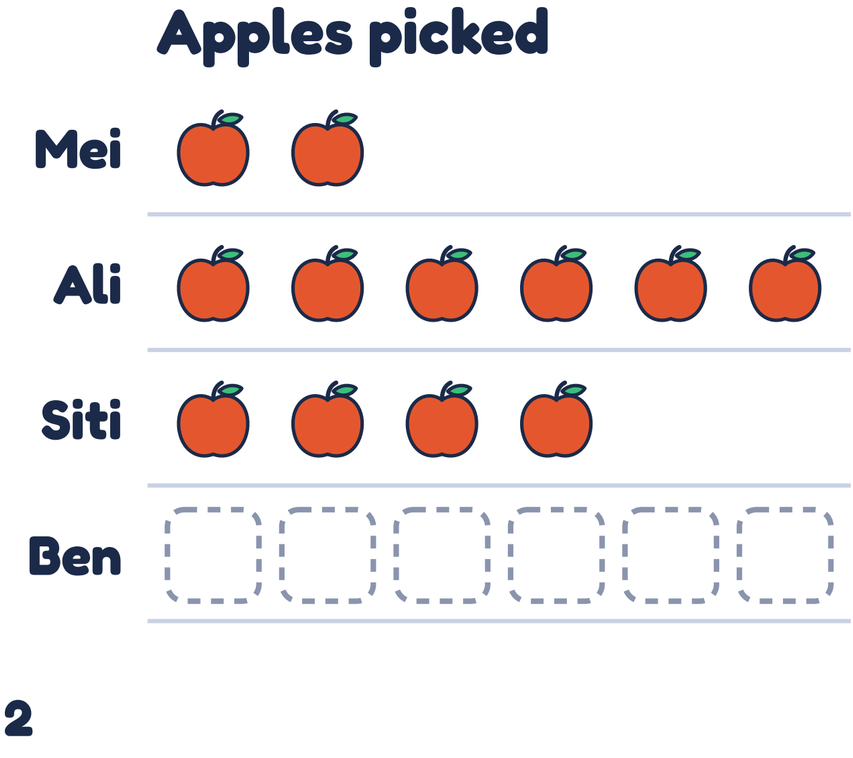
23–34 MIN




**Ali picked 12 apples: 2, 4, 6, 8, 10, 12. Ali picked 8 more apples than Mei: 4 more pictures of 2.**

# Draw with a key

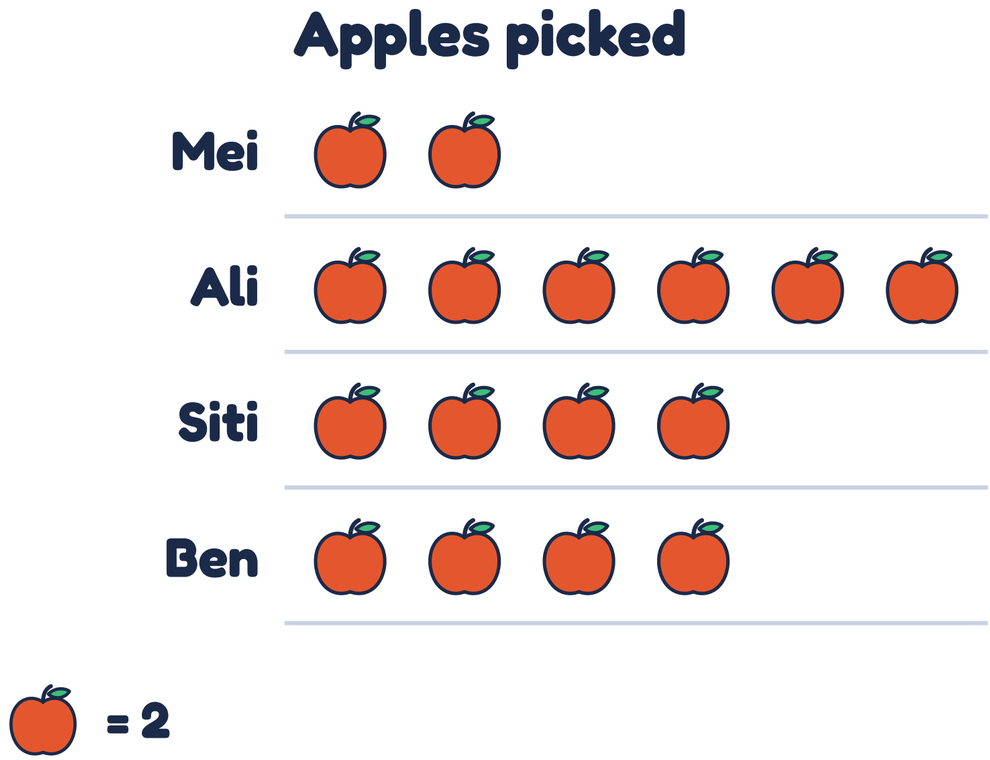
23–34 MIN




Ben picked 8 apples. Each picture stands for 2. Draw Ben's row.

# 4 pictures for 8 apples

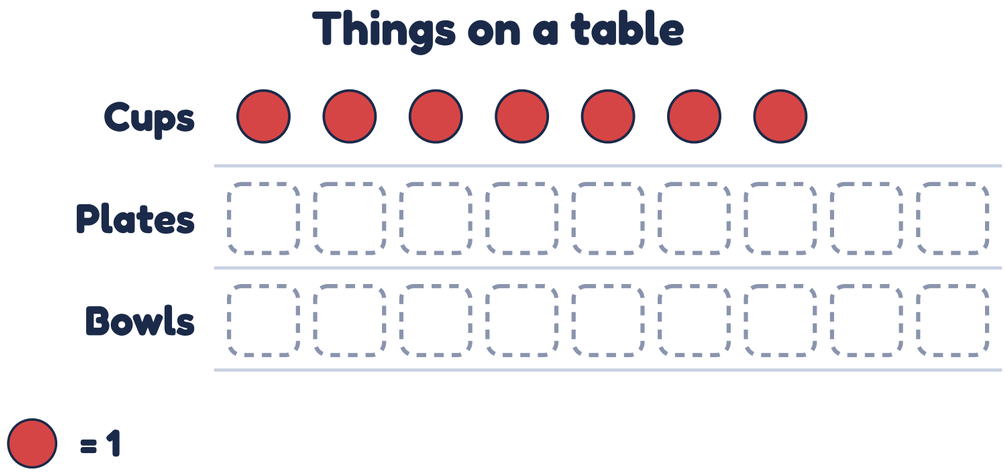
23–34 MIN




**8 apples: 2, 4, 6, 8. That is 4 pictures.**

# The missing bar

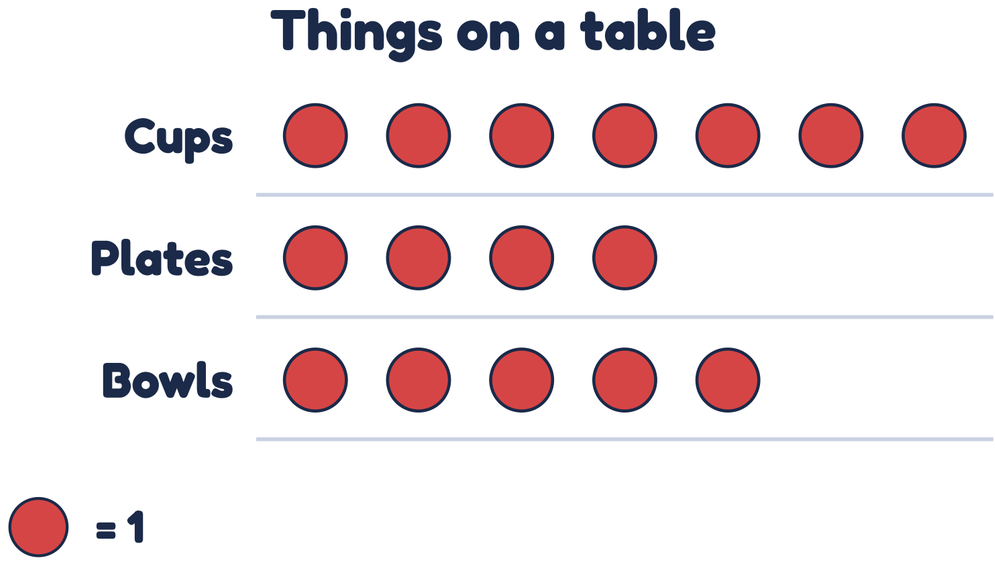
23–34 MIN




Each picture stands for 1. There are 16 things in all. There are 3 fewer plates than cups. Draw the plates and the bowls.

23–34 MIN

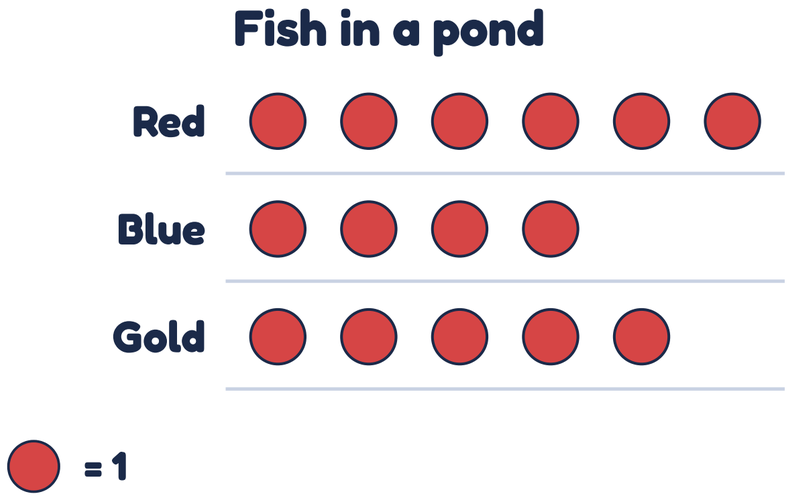
# 4 plates and 5 bowls




**7 take away 3 is 4 plates. 7 + 4 = 11. 16 take away 11 is 5 bowls. Check: 7 + 4 + 5 = 16.**

# What can a graph not tell us?

23–34 MIN




1. How many gold fish are there?
2. Which fish is the happiest?
3. How many more red fish than blue fish?

**Which question can the graph not answer? How do you know?**

# Question 2 cannot be answered

23–34 MIN

**Question 2. The graph shows how many fish of each colour there are. It does not show feelings.**

34–38 MIN

# Count and move break

Stand up. Count in 2s and jump each time.

Now count in 3s and clap each time.

Sip some water.

# Workbook page 1: Start here

38–70 MIN

Questions 1 to 3. Take your time.

**S1** Study the graph. Each picture stands for 1 book. Answer questions 1 to 3.




**S2** Look at the graph in question 1.

**S3** Look at the graph in question 1.

38–70 MIN

# Check page 1

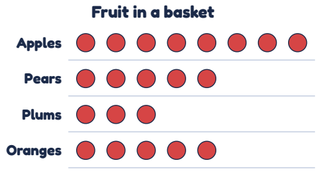
| | |
|---|---|
| S1 | Mei; Siti |
| S2 | 5 more books; 25 books in all |
| S3 | True, True, False, False |

# Workbook page 2: Start here

38–70 MIN

Questions 4 to 6. Take your time.

**S4** Study the graph. Each picture stands for 1 fruit. Answer questions 4 to 6.




**S5** Look at the graph in question 4.

**S6** Look at the graph in question 4.

38–70 MIN

# Check page 2

| | |
|---|---|
| S4 | 4 kinds; Oranges |
| S5 | 2 fewer plums than pears; 11 fruits |
| S6 | 21 fruits in all; Apples |

38–70 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | A graph from clues, a key of 2, why the longest row is not always the most |
| **Go further (page 4)** | Missing Bar, a graph from clues, which question it cannot answer |
| **Investigation (page 5)** | Look, count and ask |
| **Check** | Finish and check your own work |

**Everyone may choose. Not finished? Pages 3 to 5 are for home too.**

# Talk and explain

70–76 MIN

A CHILD IN ANOTHER CLASS SAYS

There are 21 books in all.

The first bar has 6 books and the second has 9 books.

So the third bar has 8 books.

Do you agree? How could we check?

I think … because …

# Back to the sneak-peek

70–76 MIN

**Puzzle 1: 3 pictures. Puzzle 2: 6 stickers. Each picture stands for 2, so count in 2s: 2, 4, 6.**

# Exit ticket

76–82 MIN

On your own. No looking back.

**E1** **Look at the graph. Each picture stands for 1 sticker.**

Circle the child with the fewest stickers.




**E2** **Look at the graph in E1.**

The children have ☐ stickers altogether.

**E3** **Look at the graph in E1. Mei has 3 more stickers than Ali.**

Circle True or False.

# Check the exit ticket

76–82 MIN

| | |
|---|---|
| E1 | Siti |
| E2 | 16 stickers |
| E3 | True |

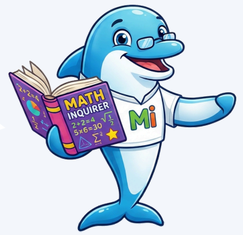
# Look-back

82–85 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ reading the key ☐ counting in 2s ☐ missing bars ☐ something else |
| **What helped me** | ☐ reading the key first ☐ counting in 2s ☐ adding to check<br>☐ I was not stuck |
| **Next time I will try** | ☐ reading the key ☐ missing bars ☐ questions a graph cannot answer<br>☐ asking a question |
| | ☐ I am still stuck (my grown-up will photograph this page for my tutor)<br>☐ I would like to pass |

**Take your time. You can pass.**



85–90 MIN

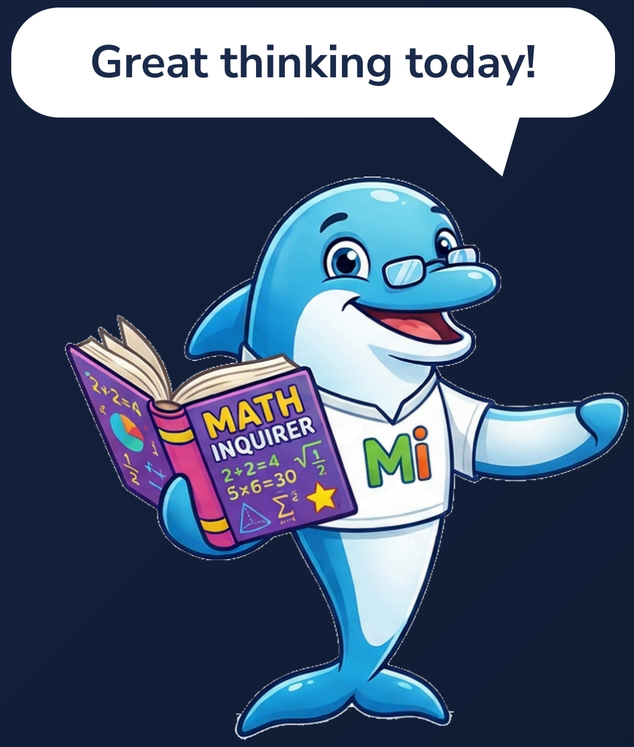
# Well done, inquirers!

Show a grown-up a graph. Say what one picture stands for.

NEXT LESSON

## The Problem-Solving Toolbox

Have ready: the tool-and-problem cards page from your book, with the 15 cards cut out by a grown-up.



# Go further: each picture is 3

OPTIONAL

## Mi's stamps

Stamps 

 = 3

Each picture stands for 3 stamps. How many stamps are in the row?

OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Is it always true?** | When every picture stands for 2, is the longest row always the most things? |
| **Make a puzzle** | Draw a graph with a key of 2 and hide one bar. Tell your grown-up the total. |
| **Teach it back** | Tell a toy why we read the key first. |

Optional. Everyone may join.